

Module Fabrication Document

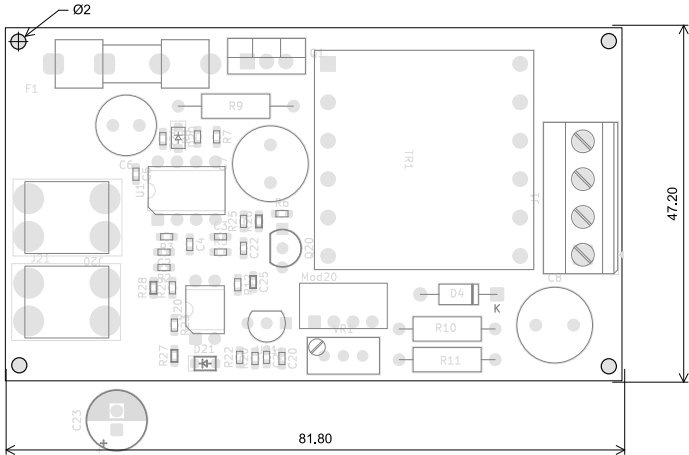
Layer Stack Legend

	Material	Layer	Thickness	Dielectric	Type	Gerber
		F,Paste			Paste Mask	
		F,Silkscreen			Legend	GBR
		F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	Copper	L1 (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
	Core		1.48mm	FR4_7628	Dielectric	
	Copper	L2 (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
		B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
		B,Silkscreen			Legend	GBR
		B,Paste			Paste Mask	
Total thickness: 1.66mm						
Note: external layer thicknesses are specified after plating						

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
NA	-	-	-	-	-	-

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR TOP: #8071501A / BOTTOM: GREEN.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
B. Tg 170 C OR EQUIVALENT.
C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY FR.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

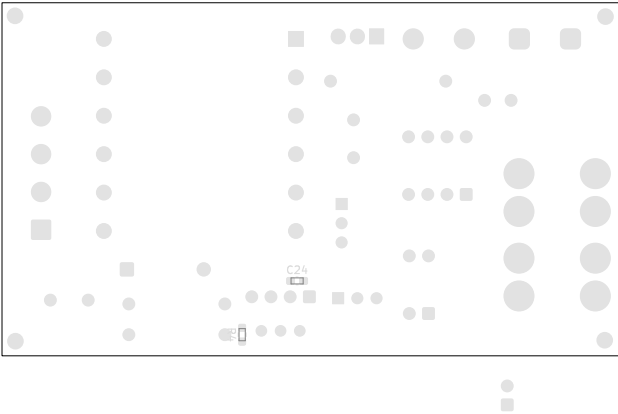
BOARD SIZE81.540 × 46.692 mm
BOARD THICKNESS1.660 mm
TRACE WIDTHN/A mm
TRACE TO TRACE0.200 mm
MIN. HOLE (PTH)0.800 mm
MIN. HOLE (NPTH)2.100 mm
ANNULAR RING0.320 mm
COPPER TO HOLE0.254 mm
COPPER TO EDGE0.250 mm
HOLE TO HOLE0.254 mm
- 11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

	Comments:	Company:		Variant:	Git Hash:
	Comment 1	FR		PRELIMINARY	47043d9
	Comment 2	Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Comment 3				
	Comment 4				
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Top Fabrication (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
	Sheet Path:		Reviewer:	Size:	Sheet:
			NA	A4	1 of 8

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Bottom Fabrication (Scale 1:1)



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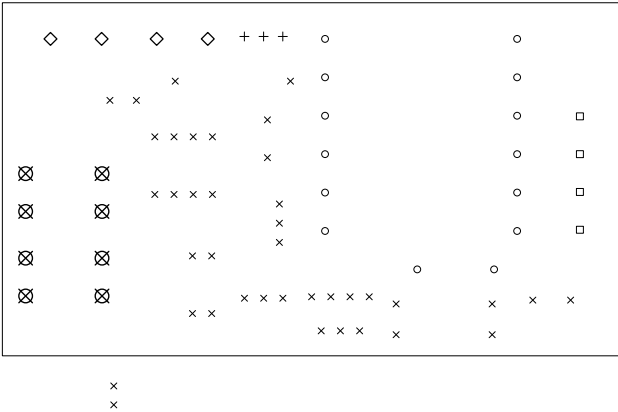
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	Comment 4	File Name:		Date:	Revision:
		Isolated_12to450V.kicad_pcb		2024-04-13	1.0.0+ (Unreleased)
	Sheet Title:	Designer:		Size:	Sheet:
	Bottom Fabrication (Scale 1:1)	FR		A4	2 of 8
	Sheet Path:	Reviewer:			
		NA			

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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	39	0,80mm (31,50mils)	PTH	Round	L1 (Sig. PWR) - L2 (Sig. PWR)	Pad
○	14	0,90mm (35,43mils)	PTH	Round	L1 (Sig. PWR) - L2 (Sig. PWR)	Pad
+	3	1,20mm (47,24mils)	PTH	Round	L1 (Sig. PWR) - L2 (Sig. PWR)	Pad
□	4	1,30mm (51,18mils)	PTH	Round	L1 (Sig. PWR) - L2 (Sig. PWR)	Pad
◇	4	1,70mm (66,93mils)	PTH	Round	L1 (Sig. PWR) - L2 (Sig. PWR)	Pad
⊗	8	1,80mm (70,87mils)	PTH	Round	L1 (Sig. PWR) - L2 (Sig. PWR)	Pad
Total 72						

Drill Drawing L1 - L2 (Scale 1:1)



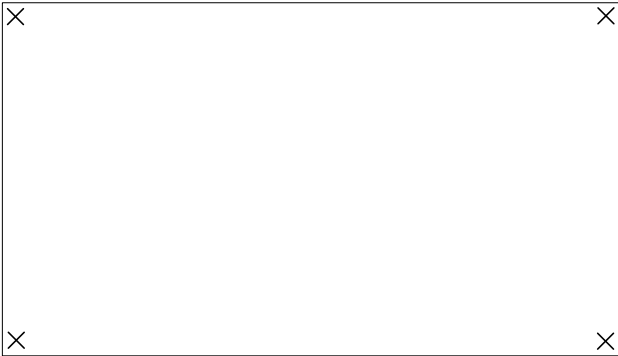
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	Comment 4	Module		Isolated step-up 12:450 V	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Drill Drawing (L1 - L2)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
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Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	2.10mm (82.68mils)	NPTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Mechanical
	Total 4					

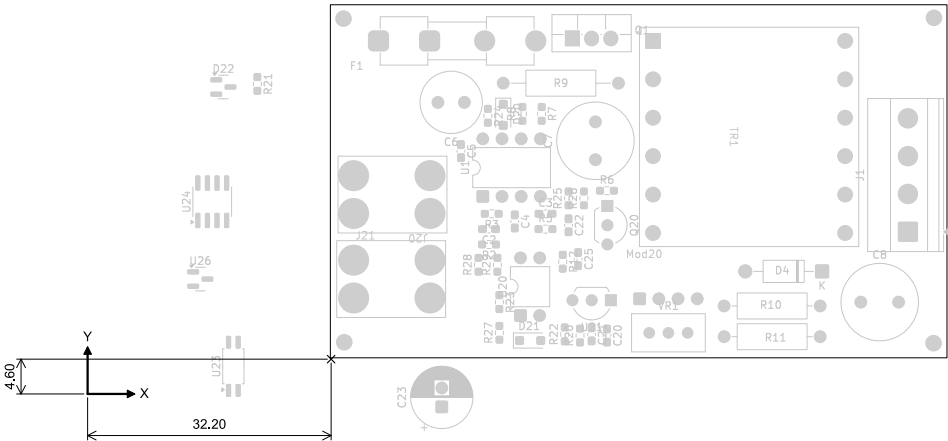
Drill Drawing L1 - L2 (Scale 1:1)



	Comments:	Company:		Variant:	Git Hash:
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	Comment 3	Module		Isolated step-up 12:450 V	
	Comment 4				
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Drill Drawing (L1 - L2)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
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Module Fabrication Document

Top Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
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Ref.	Net	X [mm]	Y [mm]
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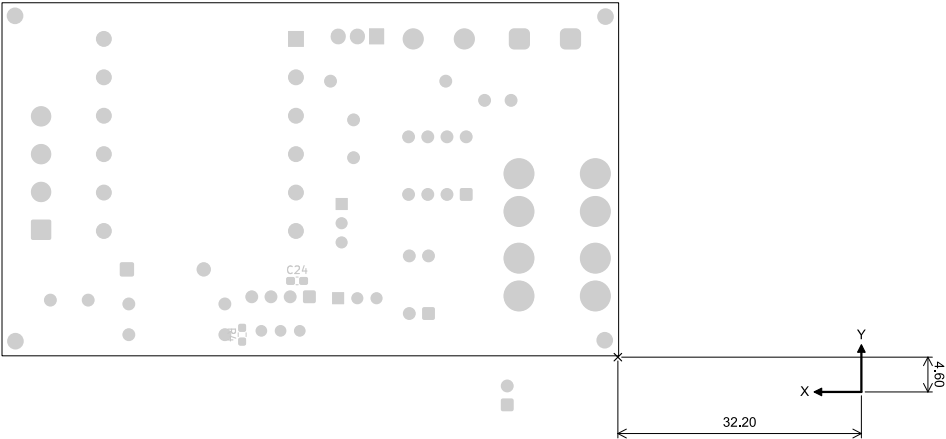
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	Comment 3				
	Comment 4	Module		Isolated step-up 12:450 V	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Top Test Points (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
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Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
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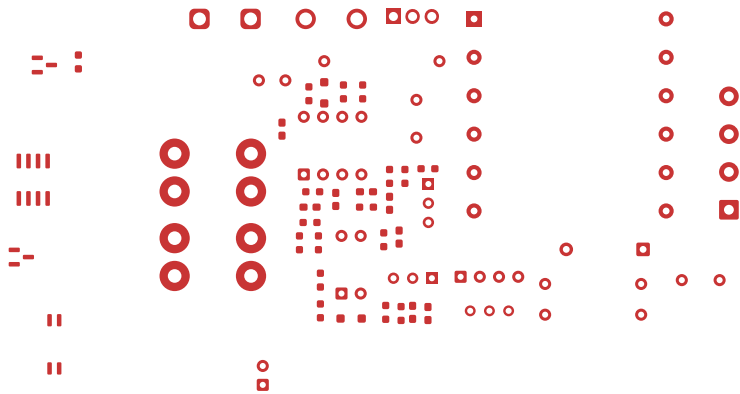


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	Comments: Comment 1 Comment 2 Comment 3 Comment 4	Company: FR		Variant: PRELIMINARY	Git Hash: 47043d9
		Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title: Bottom Test Points (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:		Reviewer: NA	Size: A4	Sheet: 6 of 8

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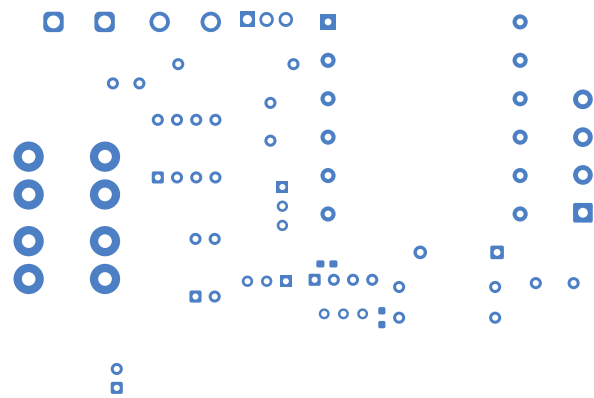
L1 (Sig, PWR) (Scale 1:1)



	Comments:	Company:		Variant:	Git Hash:
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	Comment 3			Isolated step-up 12:450 V	
	Comment 4				
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	L1 (Sig, PWR) (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
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L2 (Sig, PWR) (Scale 1:1)



	Comments:	Company:		Variant:	Git Hash:
	Comment 1	FR		PRELIMINARY	47043d9
	Comment 2	Board Name:		Project Name:	
	Comment 3				
	Comment 4	Module		Isolated step-up 12:450 V	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	L2 (Sig, PWR) (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
	Sheet Path:		Reviewer:	Size:	Sheet:
			NA	A4	8 of 8